MAT 3007 – Optimization

The Simplex Method

Lecture 08

Instructor: Junchi Yang

CUHK-Shenzhen

The Simplex Method

Simplex Method

The simplex method proceeds from one BFS (a corner point of the feasible region) to a neighboring one to continuously improve the value of the objective function until reaching optimality.

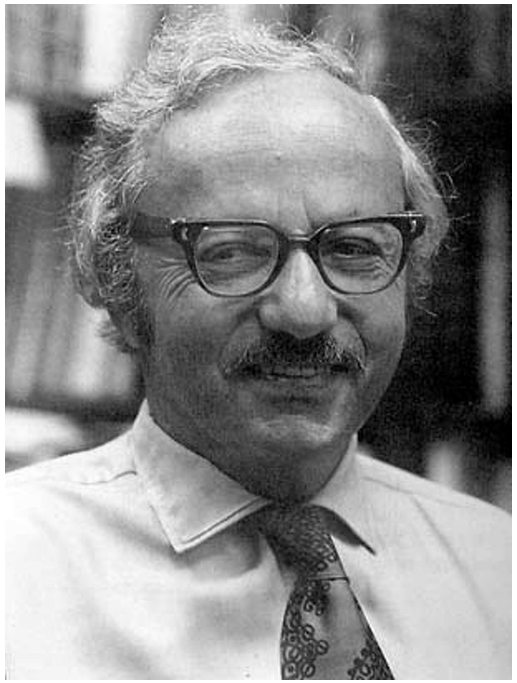


Figure 1: The simplex method was invented by George Dantzig.

Neighboring BFS

Definition: Neighboring Basic Solutions

Two basic solutions are neighboring (or adjacent) if they differ by exactly one basic (or non-basic) index.

- For example, a BFS constructed by using the columns $\{1, 2, 3\}$ is a neighbor to the BFS constructed by using the columns $\{1, 3, 5\}$ (but not $\{1, 4, 5\}$).

Finding and Moving to Neighboring BFS

Simplex Method: Starting Point

Starting from a BFS $\boldsymbol{x}$ with basis is $B(1), \ldots, B(m)$,

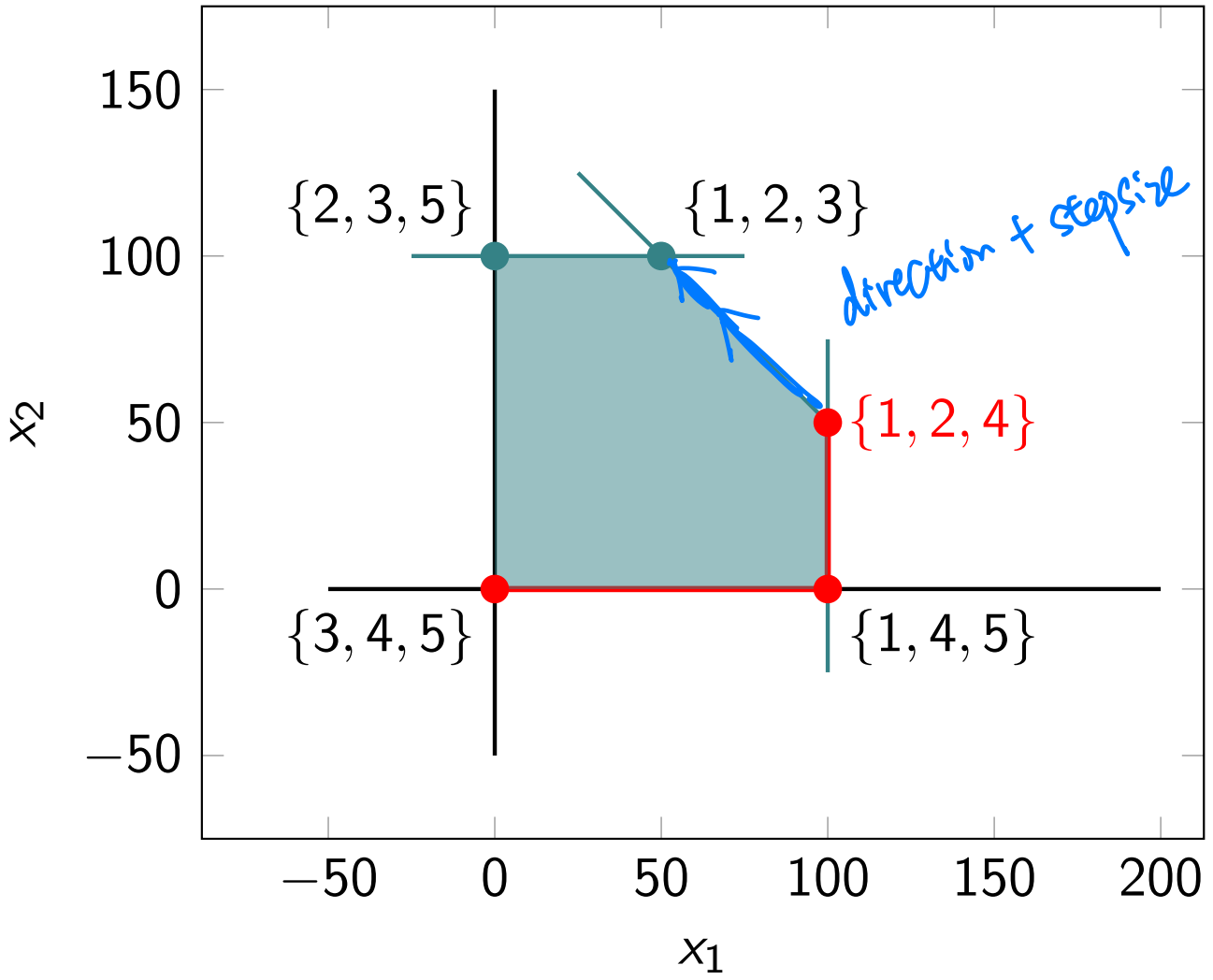
$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

Denote N the non-basic index set. B and N are disjoint, $|B| = m$, $|N| = n - m$, and $B \cup N = \{1, 2, ..., n\}$. Let $\boldsymbol{A}_N$ be the matrix consisting of the non-basic columns of $\boldsymbol{A}$.

Rearranging the variables, we can write $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, $\boldsymbol{x} = [\boldsymbol{x}_B; \boldsymbol{x}_N]$, where $\boldsymbol{x}_B$ are the basic variables, and $\boldsymbol{x}_N$ are the non-basic variables.

$$\boldsymbol{x}_B = \boldsymbol{A}_B^{-1}\boldsymbol{b} \geq \boldsymbol{0}, \qquad \boldsymbol{x}_N = 0.$$

Example


150
100
50
0
−50
x_2
−50
0
50
100
150
200
x_1
{2, 3, 5}
{1, 2, 3}
direction + stepsize
{1, 2, 4}
{3, 4, 5}
{1, 4, 5}

Finding a Neighboring BFS

We want to find a neighboring BFS:

- Finding a neighbor means changing one of the basic indices.
- We want to select a non-basic variable x_j to enter the basis.
- This means, we want to increase x_j from the current BFS.

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We want a non-basic variable x_j (with $j \in N$) to be basic. We consider moving $\boldsymbol{x}$ (the current BFS) to $\boldsymbol{x} + \theta \boldsymbol{d}$, $\theta \geq 0$, where:

stepsize direction

$\boldsymbol{d}_N$ {

1. $d_j = 1$. $\rightarrow x_j + \theta d_j = \theta$
2. $d_{j'} = 0$ for all other non-basic indices. $\rightarrow x_{j'} + \theta d_{j'} = 0$
3. How to choose $d_{j'} = 0$ for basic indices, i.e., $\boldsymbol{d}_B$?

$\boldsymbol{d}_N = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ ← 1 is at the position that corresponds to x_j.

What constraints do we have on $\boldsymbol{d}$? First, We need to guarantee that the resulting new point $\boldsymbol{x} + \theta \boldsymbol{d}$ is feasible:

$$\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b} = \boldsymbol{A}\boldsymbol{x} \quad \Longrightarrow \quad \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}.$$

new x

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Since $d_j = 1$ and $d_{j'} = 0$ for all other non-basic indices,

$$\boldsymbol{0} = \begin{bmatrix} \boldsymbol{A}_B & \boldsymbol{A}_N \end{bmatrix} \begin{pmatrix} \boldsymbol{d}_B \\ \boldsymbol{d}_N \end{pmatrix} = \boldsymbol{A}_B \boldsymbol{d}_B + \boldsymbol{A}_j. \Longrightarrow \boldsymbol{d}_B = -\boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

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This means that the direction $\boldsymbol{d}$ is uniquely determined:

$$\boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N] = [-\boldsymbol{A}_B^{-1} \boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0],$$

where the 1 is at the j-th entry. We call such $\boldsymbol{d}$ the j-th basic direction.

Basic Directions

We also need to consider the constraint $\boldsymbol{x} \geq \mathbf{0}$. We need to verify that this still holds (i.e., $\boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}$):

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- For basic variables, as long as they were all strictly positive, there must exist a small θ such that $\boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}$.

Observation:

- Typically, the basic variables of a BFS are all positive (i.e., m positive entries).
- There are cases that some basic variables in a BFS are equal to 0 (we call it "degeneracy"). We will discuss it later.

Non-degenerate = all basic variables are strictly positive.

Reduced Costs

Change in the Objective Value

The objective function of the original LP is $\boldsymbol{c}^\top \boldsymbol{x}$. We write

$$\boldsymbol{c} = [\boldsymbol{c}_B; \boldsymbol{c}_N].$$

When we move from $\boldsymbol{x}$ to $\boldsymbol{x} + \theta \boldsymbol{d}$, the change in objective function is $\theta \boldsymbol{c}^\top \boldsymbol{d}$.

$$c^T(x+\theta d) - c^T x = \theta c^T d$$

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When we move from $\boldsymbol{x}$ to $\boldsymbol{x} + \theta \boldsymbol{d}$, the change in objective function is $\theta \boldsymbol{c}^\top \boldsymbol{d}$. As $\boldsymbol{d}$ is the jth basic direction,

$$\boldsymbol{c}^\top \boldsymbol{d} = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j =: \bar{c}_j.$$

We call $\bar{c}_j$ the reduced cost of the variable x_j.

$$c^T d = (c_B^T \; c_N^T) \begin{pmatrix} d_B \\ d_N \end{pmatrix} = c_B^T d_B + c_N^T d_N$$

$$= -c_B^T A_B^{-1} A_j \qquad = c_j$$

Reduced Costs

The reduced cost

change in objective function

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j \qquad \theta \cdot \bar{c}_j$$

is a very important concept in the simplex method. It corresponds to the change of the objective value if one tries to change the basis.

- The first term is the cost per unit increase in the variable x_j.
- The second term is the cost of compensating the change in the basic variables necessitated by the constraints $\boldsymbol{Ax} = \boldsymbol{b}$.

Reduced Costs

The reduced costs are the indicators of non-basic index we want to add to the basic set:

- positive reduced cost: incorporating j into the current basis will increase the objective function (not good).
- negative reduced cost: incorporating j will reduce the objective function (we want to go in that direction).

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Question: What is the reduced cost of a basic variable? Or what is $\bar{c}_j$ when $j = B(i)$?

$$\bar{c}_{B(i)} = c_{B(i)} - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_{B(i)} = c_{B(i)} - \boldsymbol{c}_B^\top \boldsymbol{e}_i = c_{B(i)} - c_{B(i)} = 0.$$

Therefore, the reduced costs for basic variables are zero ($\boldsymbol{e}_i$ is the i-th unit vector.

Reduced Costs

Theorem: Stopping Criterion

Let $\boldsymbol{x}$ be a BFS associated with the basis $B(1), \ldots, B(m)$ and let $\bar{\boldsymbol{c}}$ be the corresponding vector of reduced costs. If $\bar{\boldsymbol{c}} \geq 0$, then $\boldsymbol{x}$ must be optimal.

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Remark:

- This theorem gives a stopping criterion for the simplex algorithm: We stop when all the reduced costs are nonnegative.
- It also means that if we can not find a neighbor solution that improves the objective function, then we must have already found an optimal solution.

Proof

Suppose at a BFS $\mathbf{x}$, all the reduced costs $\bar{\mathbf{c}} \geq 0$. Consider any feasible solution $\mathbf{y}$, and define $\mathbf{d} = \mathbf{y} - \mathbf{x}$.

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$Ad = A(y-x) = Ay - Ax = b - b = 0$

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$$\boldsymbol{A}\mathbf{d} = \boldsymbol{A}_B \mathbf{d}_B + \sum_{i \in N} A_i d_i = 0.$$

Here N is the set of non-basic indices. Thus

$$\mathbf{d}_B = -\sum_{i \in N} A_B^{-1} A_i d_i$$

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Now we study the change of objective value:

$$\mathbf{c}^T \mathbf{d} = \mathbf{c}_B^T \mathbf{d}_B + \sum_{i \in N} c_i d_i = \sum_{i \in N} (c_i - c_B^T A_B^{-1} A_i) d_i = \sum_{i \in N} \bar{c}_i d_i \geq 0$$

We know that all the $\bar{c}_i$'s are nonnegative, also all the d_i must be nonnegative, since $\mathbf{y} \geq 0$ and $x_i = 0$ for $in \in N$. Therefore, $\mathbf{c}^T \mathbf{d} \geq 0$, which means the objective value at $\mathbf{y}$ must be at least as large as the objective value at $\mathbf{x}$. □

Example: Reduced Costs

Consider the production problem in standard form:

$$
\begin{array}{rrrrrrrl}
\text{minimize} & -x_1 & -2x_2 & +0s_1 & +0s_2 & +0s_3 & & \\
\text{subject to} & x_1 & & +s_1 & & & = & 100 \\
& & 2x_2 & & +s_2 & & = & 200 \\
& x_1 & +x_2 & & & +s_3 & = & 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & \geq & 0
\end{array}
$$

If we are at basis $\{1, 2, 3\}$ with BFS $(50, 100, 50, 0, 0)$. Then the reduced costs are

$$
\bar{c}_4 = 0 - [-1, -2, 0] \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0.5
$$

Similarly, $\bar{c}_5 = 1$. Therefore, the reduced costs are all nonnegative and the BFS is optimal.

Example: Reduced Costs — Continued

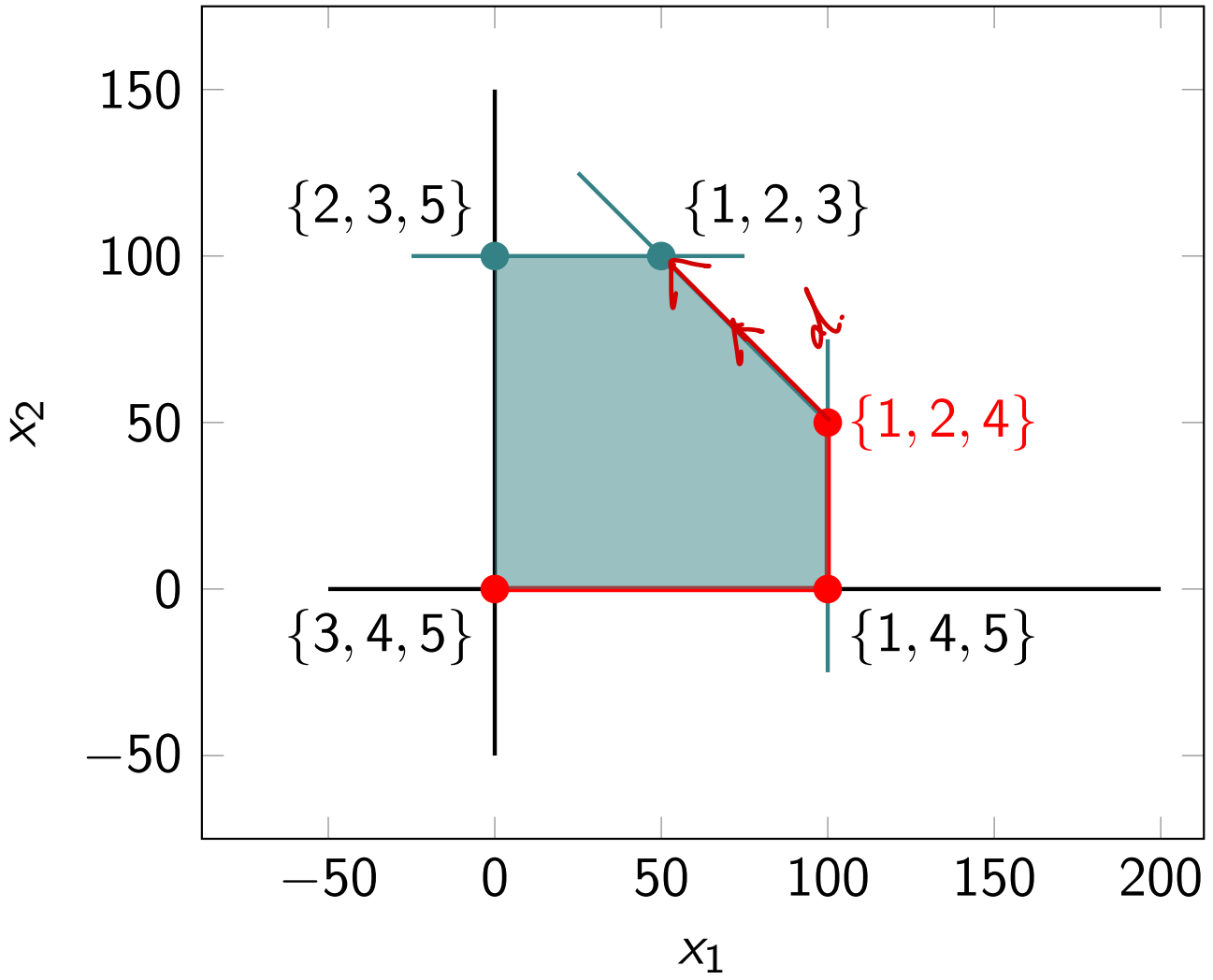
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& x_1, & x_2, & s_1, & s_2, & s_3 & & \geq 0
\end{array}
$$

If we are at basis $\{1, 4, 5\}$ with BFS $(100, 0, 0, 200, 50)$. Then the reduced costs are:

$$
\bar{c}_2 = -2 - [-1, 0, 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = -2
$$

Similarly, $\bar{c}_3 = 1$. Therefore including x_2 in the basis in the next step will reduce the objective value.

Example



Change of Basis

Observation: At a certain $\boldsymbol{x}$, if the reduced costs satisfy $\bar{c}_j \geq 0, \forall j$ then $\boldsymbol{x}$ is optimal. What if some $\bar{c}_j < 0$?

- It means that by bringing the j-th variable (non-basic) into the basis (or moving in j-th basic direction $\boldsymbol{d}$), we can decrease the objective value.

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How far can we go in the direction $\boldsymbol{d}$?

- We want θ as large as possible but still satisfy $\boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}$:

$$\theta^* = \max\{\theta \geq 0 : \boldsymbol{x} + \theta \boldsymbol{d} \geq \mathbf{0}\}.$$

Choosing a Stepsize: Minimum-Ratio-Test

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We consider two cases:

1. If $\boldsymbol{d} \geq \mathbf{0}$, then $\theta^* = \infty$. One can go unlimitedly far without making the solution infeasible, while keeping the objective decreasing. Therefore, the original LP is unbounded.

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2. If $d_i < 0$ for some i, then we can solve:

$$\theta^* = \min_{\{i : d_i < 0\}} -\frac{x_i}{d_i}.$$

Since $d_i \geq 0$ for $i \in N$, we can also write it as:

$$\theta^* = \min_{\{i \in B : d_i < 0\}} -\frac{x_i}{d_i}.$$

Moving to a New Basis

Assume θ^* is finite (otherwise the problem is unbounded), then we can move to another feasible point:

$$\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}.$$

Let $B(\ell) \in B$ be an index with $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$ (it is possible that multiple indices attain this $\rightsquigarrow$ we will discuss this later).

$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0.$$

$$\theta^* = \frac{-x_{B(\ell)}}{d_{B(\ell)}}$$

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$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0.$$

Thus, the basic variable $x_{B(\ell)}$ has become zero, whereas the non-basic variable x_j is now positive (equal to θ^*). The basis has changed to: (Are correpsonding columns of A linearly independent? Let's prove later.)

$$B(1), \ldots, B(\ell-1), j, B(\ell+1), \ldots, B(m).$$

An Iteration of the Simplex Method

Initialization: We start from a BFS $\boldsymbol{x}$ (with basic set B).

1. Compute the reduced costs $\bar{\boldsymbol{c}}$:

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2. Compute the jth basic direction

$$\boldsymbol{d} = [-\boldsymbol{A}_B^{-1} \boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0]$$

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   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$.
3. Set $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$. Then, $\boldsymbol{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. The objective function value is changed by $\theta^* \boldsymbol{c}^\top \boldsymbol{d} = \theta^* \bar{c}_j$.
4. Repeat these procedures.

Simplex Iteration: Discussion

Questions:

- Is the new point $\boldsymbol{y}$ a BFS? (We haven't check whether the columns of $\boldsymbol{A}_B$ with new basis B are linearly independent).

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- Given several j with $\bar{c}_j < 0$, which jth basic direction should we choose?
- Suppose there are multiple ℓ with $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$, how should we update the basic indices?

Theorem: Properties of y

Let $\boldsymbol{x}$ be a non-degenerate BFS ($\boldsymbol{x}_B > 0$) with basic indices B and let $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$ be generated by the simplex iteration. Then, $\boldsymbol{y}$ is a basic feasible solution associated with the basic indices

$$\bar{B} \triangleq \{B(1), \ldots, B(\ell-1), j, B(\ell+1), \ldots, B(m)\}.$$

Need to show columns of $A_{\bar{B}}$ are linearly indep.

See next lecture for proof sketch.

Simplex Iteration: Properties and Convergence

Theorem: Properties of $\boldsymbol{y}$

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Theorem: Convergence

Assume that the feasible set is nonempty and that every BFS is non-degenerate. Then, the simplex method terminates after a finite number of iterations with the following options:

1. The method stops with an BFS which is an optimal solution.
2. We have found a vector $\boldsymbol{d}$ with $\boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}$, $\boldsymbol{d} \geq \boldsymbol{0}$, and $\boldsymbol{c}^\top \boldsymbol{d} < 0$, and the optimal value is $-\infty$.

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We have seen that the objective value will strictly decrease after one simplex method iteration if the initial BFS is nondegenerate.

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However, it is possible that the objective stays the same when there are zeros in x_B.

Recall that

$$\theta^* = \min_{\{i \in B | d_i < 0\}} -\frac{x_i}{d_i}.$$

If for some $i \in B$ with $x_i = 0$ and $d_i < 0$, then we obtain $\theta^* = 0$. Hence $\boldsymbol{y} = \boldsymbol{x}$ and the objective value reamins unchanged.

Degeneracy

We call a basic feasible solution $\boldsymbol{x}$ (non-)degenerate if (none) some of the basic variables are 0.

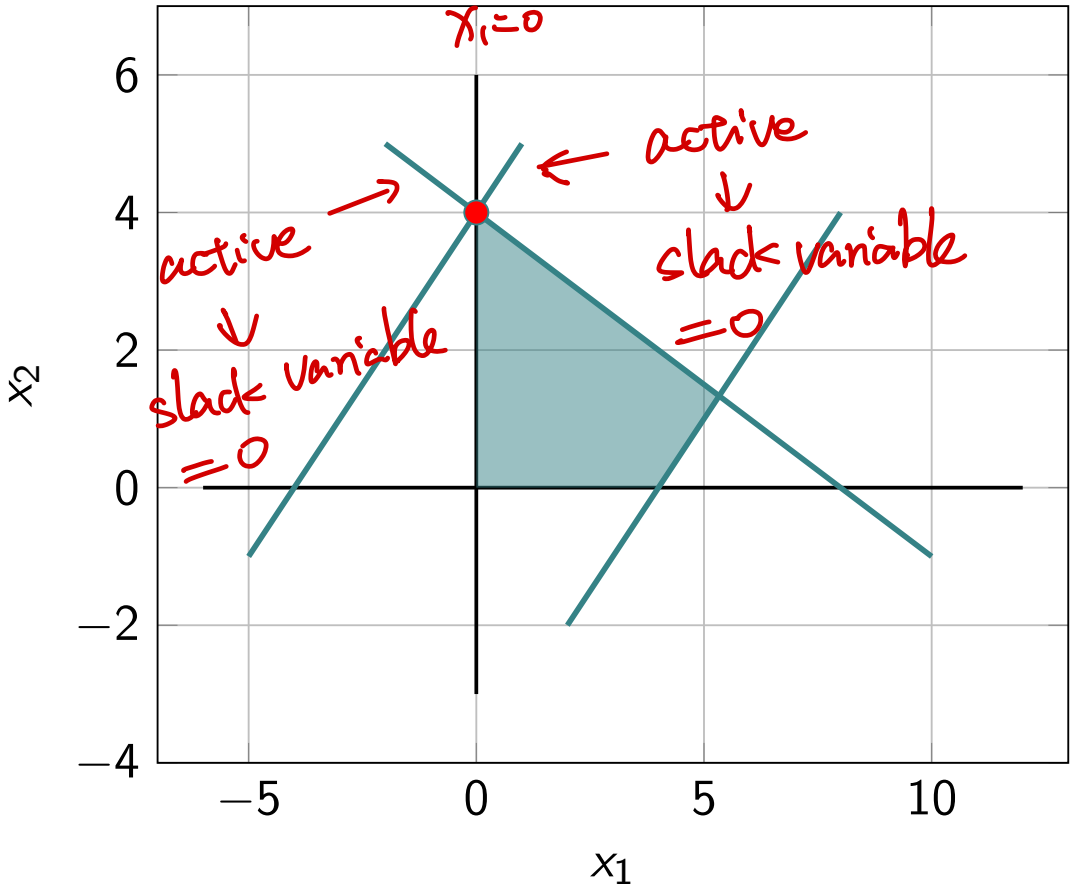
An example:

$$\begin{array}{rrrrrcl} x_1 & +2x_2 & +s_1 & & & = & 8 \\ x_1 & -x_2 & & +s_2 & & = & 4 \\ -x_1 & +x_2 & & & +s_3 & = & 4 \\ x_1, & x_2, & s_1, & s_2, & s_3 & \geq & 0 \end{array}$$

If we choose the basic indices to be $\{1, 2, 4\}$, then the corresponding basic solution is $(0, 4, 8)^\top$. It is degenerate.

$\rightsquigarrow$ The number of non-zeros at the BS is strictly less than m.

Illustration



There are 3 variables $=0$ at the red point.

- More than 2 lines intersect at one point.
- In general, at a degenerate point, more than k planes intersect at a k-dimensional space.

Degeneracy

Assume we encounter degeneracy at some point:

- Given a BFS $\boldsymbol{x}$ with negative reduced cost $\bar{c}_j < 0$ and $x_{B(\ell)} = 0$. Let $\boldsymbol{d}$ be j-th basic direction. Assume $d_{B(\ell)} < 0$ (then $\theta^* = 0$ in this case).

We can still change the basic index from $B(\ell)$ ($B(\ell)$ leaving the basis) to j (j entering the basis) and proceed to the next iteration.

- Although the iterate and the objective value do not change, the basis changes. Therefore, the reduced costs vector will change in the next iteration — issue seems resolved?

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We need to guarantee that there is not any cycle, i.e., we will not visit the same BFS more than once:

- This can only occur in case of degeneracy, since otherwise the objective value will strictly decrease!

Example of Cycling

If not dealt properly, cycling can happen. Consider the LP:

$$\boldsymbol{A} = \begin{pmatrix} -2 & -9 & 1 & 9 & 1 & 0 \\ 1/3 & 1 & -1/3 & -2 & 0 & 1 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\boldsymbol{c}^\top = (-2, -3, 1, 12, 0, 0).$$

If we set $B = \{5, 6\}$ initially, then the sequence shown below leads to a cycle (objective value doesn't change, and there is always an index with negative reduced cost):

Step #	1	2	3	4	5	6
Exiting	x_6	x_5	x_2	x_1	x_4	x_3
Entering	x_2	x_1	x_4	x_3	x_6	x_5
Basis	$\{5, 2\}$	$\{1, 2\}$	$\{1, 4\}$	$\{3, 4\}$	$\{3, 6\}$	$\{5, 6\}$

We will show that cycle can be avoided by designing how to choose incoming/outgoing basis when there are multiple choices.

Pivoting Rules: Choosing the Entering Basis

In the algorithm, we can choose an index $j \in N$ with negative reduced cost to enter the basis. Sometimes, there are more than one j with $\bar{c}_j < 0$. In this case, we need to introduce some rules to choose the entering basis.

Here are several possible rules:

$\bar{c}_2 < 0$, $\bar{c}_4 < 0$, $\bar{c}_8 < 0$
then we choose 2 to enter

1. **Smallest Index Rule**: Choose the smallest index j with $\bar{c}_j < 0$.
2. **Most Negative Rule**: Choose the smallest $\bar{c}_j$.
3. **Most Decrement Rule**: Choose j with the smallest $\theta^* \bar{c}_j$.

Pivoting Rules: Choosing the Exiting Basis

We choose an index $i \in B$ that attains the minimum

$$\theta^* = \min_{\{i \in B | d_i < 0\}} -\frac{x_i}{d_i}.$$

to leave the basis. It is possible that there is more than one index that attains the minimum (tie).Then we also need a rule to decide the exiting basis.

- The most commonly used rule is the smallest index rule.

When this tie happens, the next BFS will be degenerate. (Why?)

Bland's Rule

Theorem: Bland's Rule

If we use the smallest index rule for choosing both the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

- Proof in Bertsimas + Tsitsiklis.
- Using the Bland's rule when applying the simplex method, we can guarantee to stop within a finite number of iterations at an optimal solution.

Today's lecture

- Reduced cost $\bar{c}_j = c_j - c_B^T A_B^{-1} A_j$
- Stopping criterion
- Choice of stepsize and change of basis
- Degeneracy

Next lecture

- Initialization of the simplex method
- Simplex tableau

Questions?